

Alessio Salvatore

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Education

ETH Zürich, Zürich, Switzerland Sep 2025–Present
M.Sc. in Robotics
Coursework: Perception and Learning for Robotics, Robot Learning, 3D Vision, Probabilistic Artificial Intelligence

Politecnico di Milano, Milan, Italy Sep 2022–Jul 2025
B.Sc. in Automation Engineering
Final Grade: 110/110 *cum laude* (equivalent GPA $\approx$ 4.0/4.0; 29.57/30).

Illinois Institute of Technology, Chicago, IL, USA Aug–Dec 2024
Exchange — Global E3 Program

Projects

Research Projects

Goal-Driven Adaptive-Resolution 3D Mapping with RL  Report Feb–Jun 2026
Computer Vision and Geometry Group – Marc Pollefeys Lab

- Trained a PPO policy that adapts per-region voxel resolution (split/keep/merge) online during navigation under a hard memory budget, reformulating an intractable per-block action space into a compact two-threshold policy
- Re-implemented PEANUT's dense voxel map as a sparse hierarchical multi-resolution grid (Habitat, HM3D); the policy retains 79% of point-goal success with 87% less memory than a uniformly fine map (20 unseen episodes)

Knowledge Distillation for Aerial Robots Control  Video  Report Feb–Jun 2026
Autonomous Systems Lab – Roland Siegwart Lab

- Distilled model-based controllers for an overactuated tilt-rotor MAV (OMAV) into a single recurrent LSTM policy, trained closed-loop through differentiable dynamics with DAGger on ~ 42 h of simulated flight
- In simulation the policy outperforms the nominal model-based controller in velocity tracking ($p < 10^{-10}$, 2,000 paired rollouts) and nearly halves divergence under $\pm 30\%$ model mismatch; deployed zero-shot on the real OMAV

Selected Projects

Autonomous Exploration on a Unitree A2 Quadruped  Repo 06/2026
ETH Robotics Summer School (ETHZ-RobotX)

- Deployed a full autonomous search-and-explore mission with our team on a real Unitree A2 quadruped across three field sites; developed the artifact-mapping node fusing YOLO detections into a deduplicated map-frame object log

Vision-Language-Action Policies on a Real SO-101 Arm  Repo  Video 05/2026
Project of the Class Robot Learning

- Fine-tuned SmoVLA policies for language-conditioned pick-and-place across three tasks with on-robot inference at 10 FPS; built the LeRobot data pipeline and automated H100 cloud training

Person Following on the Agibot X2 Humanoid  Repo  Video 05/2026
RoboHack Hackathon

- Built real-time person following running fully onboard in a hackathon weekend: YOLOv8 on the head stereo cameras with disparity-based depth, Ruckig head tracking, and a state-machine locomotion supervisor holding a set distance

Work Experience

JOINT Lab — Italian Institute of Technology, Bergamo Feb–Jun 2025
Research Intern — Industrial Robotics  Repo

- Designed a linear actuator in CAD from commercial off-the-shelf components, implemented real-time control on Arduino, and validated system dynamics with a MATLAB simulation

ETH Robotics Club Nov 2025–Present
Partnerships Team Lead

- Secured sponsorships for club initiatives and events

Laelaps AI Jun 2026–Present
Robotics Engineer

- Developing multi-agent systems for robot orchestration in complex real-world environments